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From N64 Patch to CI Gate: An AI-Orchestrated Pipeline for Retro Game Development

Contract Hardening: Implementing Machine-Checkable Invariants for the N64 Slice

The foundational objective of deepening the Nintendo 64 (N64) vertical slice is to transform high-level design principles into a set of rigorously enforced, machine-checkable invariants. This process involves moving from abstract concepts of "safe patching" and "budget management" to concrete, deterministic validators written in Rust. These invariants form the bedrock of the Sonia framework's promise of a contract-first development environment. The strategy prioritizes the implementation of four critical invariant modules: patch safety, hardware budgets, determinism, and Application Binary Interface (ABI) stability. Each of these modules must be designed not only to pass or fail a given artifact but also to provide rich, structured diagnostic information that enables a constructive feedback loop with the AI agent. The implementation of these checkers within the `ai_checklist.rs` orchestrator creates a composable pre-flight validation matrix, ensuring that every piece of AI-generated code or data adheres to the strict contractual boundaries of the N64 platform before it can proceed further in the pipeline.

The first and most critical invariant to implement is **patch safety**, which is formally defined by two core principles: interval containment and interval disjointness . This concept treats the N64 ROM as a linear memory map where segments have fixed start addresses and sizes [33] [34]. A patch operation, such as replacing a texture file or modifying game code, can be represented as an interval on this map. The first rule, interval containment, mandates that any newly generated patch interval must be entirely contained within a designated "mutable" segment of the ROM. This prevents writes to read-only regions like bootcode or critical assets, which could lead to catastrophic failure [46]. The second rule, interval disjointness, requires that no two active patches may overlap in their byte ranges. Violating this would result in ambiguous or corrupted data. The implementation of this invariant can be achieved through a lightweight checker that maintains a list or tree of existing patch intervals. When a new patch proposal is made, the checker verifies that its proposed interval does not intersect with any existing one and

lies completely within the bounds of a mutable segment. The output of this check must be a structured violation report, ideally including metadata such as the line and column number of the offending `PatchSpec` definition, to provide the AI with a clear path to correction .

The second major invariant is the enforcement of **hardware budgets**. The N64 has well-defined physical limitations, including 4 MiB of RDRAM expandable to 8 MiB, which serves as both system RAM and VRAM [46]. Managing these resources is paramount to creating stable games. The research blueprint proposes modeling these constraints as a system of linear inequalities over various resource pools . For instance, the total size of the ROM must be less than or equal to the cartridge capacity ceiling ($\sum_i s_i^{\text{rom}} \leq C_{\text{rom}}$), and the combined VRAM usage of all textures must not exceed the available texture pool ($\sum_{i \in \text{Textures}} s_i^{\text{vram}} \leq C_{\text{tex}}$) . The implementation in `crates/sonia-core/src/invariants/hardware_budget.rs` must parse a `RomLayout`, which contains size information for each asset, and compare it against a `N64Constraints` structure defining the platform's limits. Crucially, the checker should not simply return a boolean pass/fail. Instead, it must compute granular budget deltas and overflows, providing detailed diagnostics like "textures +4 MB over budget" . This transforms a hard failure into a productive interaction. The AI, upon receiving such a report, can then be prompted to propose trade-offs, such as down-sampling certain textures or compressing audio assets, with quantified savings estimates provided by the validator itself .

Third, **determinism** is a non-negotiable requirement for many types of games, especially those involving multiplayer netcode or save/replay functionality. The research identifies several forbidden patterns, such as the use of `rand::thread_rng` or `std::time`, as sources of non-determinism . The implementation in `crates/sonia-core/src/invariants/determinism.rs` should begin with a heuristic scanner that inspects source code for these forbidden imports and function signatures. More advanced implementations could involve parsing the Abstract Syntax Tree (AST) to detect more subtle violations, such as iterating over a `HashMap` without a fixed hasher, which leads to non-deterministic orderings across different runs [42]. The power of this invariant lies in linking static analysis to dynamic testing. If a crate fails a dedicated determinism replay test harness, the error report should explicitly reference the static rules that were violated, providing a complete audit trail for the AI . The "Determinism Checklist Autocorrection" feature requires that the violation report include structured suggestions for fixes, such as recommending the replacement of a `HashMap` with an `IndexMap` .

Finally, **ABI stability** must be protected to ensure backward compatibility with frontends and other engines built on top of the core engine. The `crates/sonia-core/src/invariants/abi_guard.rs` module acts as the gatekeeper for C API changes. It

should operate by diffing the current C API surface, extracted using tools like `bindgen` or `cbindgen`, against a baseline stored in version control . Any breaking change to exported symbols—such as altering a function's signature, changing the layout of a `GM_*` struct, or removing an enum variant—should trigger a hard failure. To make this actionable for an AI, the protocol must support specific, machine-readable error codes for common ABI mistakes, such as `ABI_Breaking_Change_Public_Symbol`. This allows the AI to follow a documented retry path, such as generating a migration wrapper or bumping the version number, rather than failing silently or requiring manual intervention [7]. The AI-exclusive feature here is the ability to be blocked from making modifications that break public contracts without an explicit flag and accompanying migration script, ensuring long-term stability .

These four invariant modules—the patch safety checker, the hardware budget calculator, the determinism scanner, and the ABI guard—are not standalone entities. They are designed to be orchestrated by the `ai_checklist.rs` module. This central component defines a `Check` trait and implements a matrix that can run all these validators in sequence against an AI-generated artifact bundle . The output is a comprehensive pass/fail matrix that details which checks passed, which failed, and provides links to the rich diagnostic reports from each individual checker. This matrix becomes the ultimate verdict before any artifact is committed to disk or passed to a downstream tool like Starzip. By building this layer of composable, machine-checked invariants, the N64 vertical slice moves beyond a simple scripting interface and becomes a true, contract-enforcing platform, capable of guiding an AI toward safe and correct modifications within the complex constraints of classic console hardware.

Invariant Module	Core Principle	Mathematical Formulation / Concept	Key Output for AI
Patch Safety	Interval Containment & Disjointness	$[o_i, o_i+\ell_i] \subseteq [s_j, s_j+L_j]$ and $[o_i, o_i+\ell_i] \cap [o_k, o_k+\ell_k] = \emptyset$ for $i \neq k$	Structured violation report with conflicting patch intervals and affected segment details.
Hardware Budgets	Linear Inequality Systems	$\sum_i s_i^{\text{resource}} \leq C_{\text{resource}}$ for each resource (ROM, VRAM, CPU)	Granular budget delta report showing over-budget dimensions and potential trade-off suggestions.
Determinism	Forbidden Pattern Detection	Logical predicates over program features (imports, collection types, APIs). Example: $\neg(\text{use}(\text{import}=\text{"}rand::thread_rng\text{"}))$	List of violated rules and structured suggestions for refactoring (e.g., replace HashMap with IndexMap).
ABI Guard	Baseline Diffing	Set difference between current and historical ABI symbol sets, with structural equality checks.	Breaking change report with symbol names, old/new signatures, and required actions (bump version, write migration).

This systematic approach to implementing and integrating these invariants ensures that the N64 slice is not just functional but also robust and safe. The resulting system provides

the AI with immediate, actionable feedback, dramatically reducing the rejection rate of invalid proposals and fostering a more productive and efficient development loop .

Session Intelligence: Establishing the SessionProfile as the AI Orchestrator Authority

A pivotal aspect of the research goal is to shift the paradigm of AI orchestration from a reactive, command-driven model to a proactive, context-aware one. The proposed solution centers on elevating the `SessionProfile` from a simple state container to the supreme authority governing the AI's capabilities. This strategy directly addresses the user's directive to expose "orchestra powers" primarily through expanded session conditioning, positioning CI lanes as secondary enforcement layers rather than the primary orchestration surface . By enriching the `SessionProfile` with per-platform invariants, real-time CI status, and specialized execution modes, the system can dynamically adapt the AI's behavior based on the project's current needs and constraints. This transforms the AI from a tool that executes commands into a collaborator that understands the project's goals and operates within them.

The cornerstone of this intelligence model is the enrichment of the `SessionProfile` structure itself. The blueprint suggests adding specific fields for each target platform, such as `n64_rom_ceiling`, `ps1_iso_budget`, and flags like `allow_non_deterministic_experiments` . These fields act as hard-coded rulesets that define the operational boundaries for the AI when working on a particular platform. For example, if `n64_rom_ceiling` is set to 32 MiB, the AI, guided by its conditioning prompts, should automatically avoid proposals that would increase the ROM size beyond this limit. This makes the `SessionProfile` the single source of truth for all active constraints, preventing ambiguity and ensuring that the AI's proposals are always aligned with the project's fundamental requirements . The enforcement of these rules is handled by the CLI's gating logic, which uses a predicate like $allowed(c,S)$, where c is a command and S is the session profile, to reject any operation that violates the active invariants .

To make the `SessionProfile` truly intelligent, it must be continuously updated with real-time feedback from the project's health metrics. This is the role of the CI digest integration mechanism, centered around `crates/sonia-core/src/session/ci_digest.rs` . The CI pipeline, upon completion, parses the logs from various stages —including `cargo test`, Clippy linting, schema validation, and emulator smoke tests— and translates them into a structured `CiFailure` object . Each `CiFailure` entry carries

metadata such as the crate involved, the nature of the failure (e.g., `BudgetOverflowN64`, `DeterminismViolationCoreEcs`), and a descriptive message . These objects are then aggregated and stored in the `ciStatus` field of the `SessionProfile` via the `gamemodeai-session update-ci-status` command . This transforms the session state into a living document that reflects the current state of the build and test suite.

With this enriched `SessionProfile`, the AI can be conditioned to perform sophisticated, context-aware reasoning. The `ai_prompts/sonia-system-prompt.md` serves as the template that guides the AI's behavior, but instead of being a static document, it can be populated with the current session's state . Before generating a response, the AI queries the session state. If the `ciStatus` indicates recent failures related to N64 budget overflows, the AI can automatically enter a "fix-only" mode, restricting its proposals to optimization tasks like asset compression or simplification . Conversely, if all tests are passing, it might switch to a more exploratory mode, `explore_layouts`, allowing for more creative or experimental changes. This dynamic adjustment of the AI's mode of operation based on empirical evidence from the CI system is a powerful application of session intelligence. The `compute-mode` field, an AI-exclusive addition to the `SessionProfile`, can codify these behaviors, with the CLI enforcing that certain commands are disallowed in restrictive modes like `fix_only` .

A significant extension to this model is the introduction of a constraint satisfaction scoring function, as outlined in the research objectives for `session/conditioning.rs` . Instead of a binary pass/fail evaluation, each AI proposal is assigned a score based on how well it satisfies the weighted constraints defined in the `SessionProfile`. Hard constraints (e.g., staying within ROM budget, maintaining determinism) carry a large penalty for violation, potentially leading to an infinite negative score and immediate rejection. Soft constraints (e.g., adhering to coding style, minimizing code churn) contribute positively to the score. The overall score can be calculated as a weighted sum:

$$\text{score}=\sum_{i=1}^{n} w_i s_i$$

where w_i is the weight of constraint i and s_i is an indicator of its satisfaction . This allows the AI to explore a space of acceptable solutions and choose the one with the highest score, effectively optimizing for the project's priorities. The AI can even request a "dry-run" score for a proposed change, allowing it to refine its idea before committing to any disk writes . This turns the AI into an optimization engine that actively seeks to improve the codebase according to the established invariants. The entire system is

grounded in logical formulas and mathematical models, providing a provably sound foundation for AI-driven development within the rigid constraints of retro gaming hardware .

End-to-End Automation: Designing the N64 CI Pipeline and Artifact Bridge

The successful validation of the N64 vertical slice hinges on the creation of a seamless, automated, and reproducible end-to-end pipeline. This pipeline serves as the crucible in which all the preceding architectural decisions—contract hardening, session intelligence, and invariant enforcement—are forged into a functional reality. The two primary components driving this automation are the `sonia_bridge.rs` module, which acts as a semantic adaptor between high-level concepts and low-level artifacts, and the `.github/workflows/sonia-ai-n64-slice.yml` CI workflow, which functions as the hard gate and feedback loop. Together, they establish the definitive "green path" from an AI-generated proposal to a validated, patched ROM, proving the viability of the entire Sonia model for the N64 platform.

The `crates/n64-layout/src/sonia_bridge.rs` module is the linchpin of the entire vertical slice, tasked with proving the seamless integration of the N64-specific `RomLayout` and `PatchSpec` types with the universal `ArtifactSpec` format of the Sonia framework . Its responsibilities are multifaceted. First and foremost, it must provide robust conversion utilities, such as an `impl From<RomLayout> for ArtifactSpec` . This function would take a structured representation of the desired ROM layout and translate it into the canonical JSON payload expected by the rest of the Sonia ecosystem. Second, the bridge must embed validation hooks directly into its conversion logic. Before an artifact is finalized, the bridge should invoke the `ai_checklist.rs` to run a full suite of checks, including the newly implemented patch safety, hardware budget, and determinism validators . This ensures that any artifact materialized by the system is guaranteed to be compliant with all active invariants at the point of creation. Third, the bridge needs to provide encoding utilities for the payloads referenced within a `PatchSpec`, handling formats like text, hex, or base64 and correctly associating them with their corresponding files in the layout .

An advanced capability for the bridge is "patch impact introspection." Beyond simply converting a patch, the bridge could compute the precise impact region of a proposed `PatchSpec` on the final ROM. This involves analyzing the patch's operations to

determine exactly which ROM segments, memory pools, and asset classes (e.g., textures, audio, mission scripts) are affected . This structured impact summary serves two crucial purposes. For the CI system, it enables more targeted reporting; instead of a generic "patch failed," the log can state "this patch touched core rendering assets, causing a VRAM overflow." For the AI, this same information provides a retro-accurate sense of risk, allowing it to reason about the consequences of its changes before committing them . This transforms the AI from a blind code generator into a strategic planner that understands the interconnectedness of game assets.

Parallel to the bridge's development, the `.github/workflows/sonia-ai-n64-slice.yml` file must be constructed to embody the principles of a fully automated and self-contained verification loop . This CI workflow is not merely a series of disconnected jobs; it is a directed acyclic graph (DAG) of pure functions that collectively prove the validity of an AI-driven change . The workflow begins by running `tools/schema-gen/main.rs` to ensure the latest schemas are in effect for the entire validation process . The next step is to execute `sonia-core validate` against the AI-generated `ArtifactSpec`s, which triggers the full `ai_checklist` . Only upon passing this initial gate are the artifacts written to disk.

The subsequent steps represent the mechanical heart of the vertical slice. The workflow calls `starzip-cli patch` with the validated `PatchSpec` to generate a new, modified ROM image . Immediately following this, a headless emulator (such as Conk64, which supports headless mode for server-side instances [5]) is used to perform a "smoke test" on the patched ROM [1]. This test verifies that the ROM boots successfully and perhaps reaches a known checkpoint, providing a basic guarantee of functionality. The final step in the pipeline is the most critical for closing the feedback loop: parsing the results of the Starzip and emulator runs and injecting this structured digest back into the `SessionProfile` . A dedicated GitHub Actions step would call `gamemodeai-session update-ci-status`, populating the `ciStatus` field with entries for any failures, such as a `BudgetOverflow` or an `EmulatorCrash`. This completes the cycle, turning the CI pipeline from a passive verifier into an active participant in the AI's learning process. The AI, by querying the session state, gains direct, empirical knowledge of the real-world consequences of its proposals, enabling it to learn and improve with each iteration.

Mathematical Substrates and Practical Validation

The research directive emphasizes treating mathematical formalizations as reusable substrates to be validated through implementation, rather than as abstract theories to be proven in isolation . This pragmatic approach is essential for grounding the ambitious goals of the Sonia framework in the tangible realities of retro game development. The core mathematical concepts underpinning the N64 vertical slice—interval arithmetic for patch safety and polyhedral geometry for budgeting—are powerful abstractions that, when translated into deterministic Rust code, become reliable tools for enforcing platform constraints. The validation of these tools relies on a combination of unit tests, property-based fuzzing, and integration with the CI pipeline, which together provide a multi-layered defense against errors and ensure the correctness of the implementation.

The concept of **patch safety** is fundamentally rooted in one-dimensional interval arithmetic. A ROM segment can be represented as a closed interval $[S,E]$ on the address line, where S is the start address and E is the end address. A patch operation, which modifies a specific range of bytes, also corresponds to a closed interval $[o,o+l]$, where o is the offset within the segment and l is the length of the data being written. The two core safety rules translate directly into mathematical constraints: 1. **Containment:** The patch interval $[o,o+l]$ must be entirely contained within the mutable segment interval $[S,E]$. This is mathematically expressed as $S \leq o$ and $o+l \leq E$. 2. **Disjointness:** For any two distinct patches i and j, their respective intervals $[o_i,o_i+l_i]$ and $[o_j,o_j+l_j]$ must be pairwise disjoint. Two closed intervals are disjoint if and only if one ends before the other starts, i.e., $(o_i+l_i \leq o_j) \vee (o_j+l_j \leq o_i)$.

These constraints form the basis of the `Safe Patch Synthesizer` described in the research. The implementation in Rust does not need to solve a complex packing problem; it only needs to check that a proposed patch set satisfies all these linear inequalities. The "proof" of correctness comes from writing unit tests that cover edge cases (e.g., patches touching the boundary of a segment, patches of zero length) and, more importantly, from using a fuzzing framework like LibAFL to generate thousands of random patch sets, both valid and invalid, to stress-test the checker's logic [41]. The fuzzer can systematically verify that the checker rejects all violating configurations and accepts all valid ones, providing statistical confidence in its reliability [42].

Similarly, the management of **hardware budgets** is best understood through the lens of polyhedral geometry. The N64's resource constraints—ROM size, VRAM for textures and audio, runtime RAM, and CPU cycles—define a feasible region in a multi-dimensional space. Each asset in the game contributes a vector of resource consumption (e.g., a

texture might consume `[+1024, +512, 0, +1]` bytes in ROM, VRAM, and frames per second). The total resource consumption of the project is the sum of all these vectors. The budget invariants define a convex polytope (a higher-dimensional polygon) within which the total resource vector must lie. The inequalities for the N64 slice can be expressed as a system $A\mathbf{x} \leq \mathbf{b}$, where $\mathbf{x}$ is the vector of total resource usage, $\mathbf{b}$ is the vector of budget ceilings, and A is an identity matrix for simple constraints .

While solving a full mixed-integer optimization problem to find Pareto-optimal solutions is a future direction, the current implementation focuses on validation and suggestion . The `hardware_budget.rs` checker computes the left-hand side of the inequality system, $A\mathbf{x}$, and compares it against $\mathbf{b}$. If the result is positive for any dimension, a budget has been exceeded. The mathematical substrate here is the calculation of slack variables: $\text{slack}_k = b_k - (A\mathbf{x})_k$. A negative slack indicates an overflow. The "practical validation" occurs when this checker is integrated into the CI pipeline. The `starzip-budget` tool already produces a `BudgetReport` and a boolean `is_within_budget` flag . By incorporating this tool's output into the CI job and asserting its correctness on known good and bad layouts, the system gains empirical proof that the mathematical model accurately reflects the hardware's limitations. Furthermore, the checker's ability to suggest tradeoffs can be validated by simulating asset reduction scenarios and verifying that the predicted savings in the budget report match the actual changes in the `BudgetReport`.

By focusing on implementation and empirical validation, the team avoids getting bogged down in the complexities of formal proofs while still leveraging the power of mathematical abstractions. The interval arithmetic and linear inequalities provide a precise language for describing the problem domain, and the Rust code provides a deterministic, verifiable implementation. The use of modern software engineering practices like property-based fuzzing and structured CI feedback loops serves as the ultimate validation mechanism, ensuring that the mathematical substrates translate into a robust and reliable system for AI-driven development.

Strategic Roadmap: From N64 Stability to Cross-Platform Extension

The overarching strategic roadmap for the Sonia AI Model is deliberately phased, prioritizing depth and stability over premature breadth. The research goal is unequivocal: the Nintendo 64 vertical slice must be completed, hardened, and proven to be a fully

validated, end-to-end pipeline before any attempt is made to generalize the pattern to other platforms like the NES or PS1 . This "deepen before extend" principle is the central tenet of the project's growth strategy, ensuring that the core contract, invariant system, and orchestration logic are battle-tested and proven to work reliably within the unique constraints of a single platform before being applied elsewhere. The success of this phase will create a template—a "vertical slice"—that can be systematically cloned and adapted for other consoles, drastically reducing the complexity and risk associated with supporting multiple hardware architectures.

The initial phase of the roadmap is dedicated to achieving a fully green, end-to-end N64 lane. This journey begins with the foundational work on contracts and invariants. The implementation of concrete Rust validators for patch safety, hardware budgets, determinism, and ABI stability forms the first milestone . These checkers must be integrated into the `ai_checklist` and proven through a combination of unit tests and fuzzing campaigns to ensure their correctness. Concurrently, the `sonia_bridge.rs` module is developed to serve as the semantic glue connecting the N64-specific `RomLayout` and `PatchSpec` concepts to the universal `ArtifactSpec` format. This bridge is responsible for performing the final validation before any artifacts are written to disk. The culmination of this phase is the construction of the `.github/workflows/sonia-ai-n64-slice.yml` CI workflow. This workflow automates the entire green path: from schema generation and validation, through Starzip-based patching, to a headless emulator smoke test, and finally, the injection of a structured CI digest back into the `SessionProfile` . Successfully landing a pull request that adds a new N64-safe patch and passes this CI pipeline marks a critical inflection point, demonstrating that the entire system works as intended.

Once the N64 slice is deemed stable and validated, the focus shifts to solidifying the core principles of the Sonia framework. This involves documenting the contracts, workflows, and invariant rules in the `docs/sonia-ai-model.md` file, transforming it into a comprehensive "flight manual" for both human developers and the AI agent . This documentation serves as the canonical reference for the pattern, ensuring consistency and preventing drift as the system evolves. It should detail the exact sequence of CLI commands, the structure of the JSON payloads, and the specific invariants that apply to the N64 platform. At this stage, the "orchestra powers" are fully realized through the expanded session conditioning, where the `SessionProfile` acts as the authoritative gatekeeper, dictating which N64 operations the AI may perform based on the project's current state and CI health . The CI lanes now serve as a secondary UI, offering specific execution modes (e.g., `n64-budget-tuning`) that the AI can target by setting appropriate invariants in the session profile.

Only after the N64 slice is a proven, reusable pattern should the identical Sonia structure be extended to the NES. The extension process is not one of reinventing the wheel but of cloning the validated N64 structure and adapting it to the new platform's constraints. This involves creating new crates and modules for the NES-specific invariants (e.g., CHR bank limitations, PPU constraints) and updating the `hardware_budget.rs` checker to handle the NES's unique budget profiles. The `sonia_bridge.rs` would be forked to handle NES-specific payload formats and patching logic. The CI workflow would be cloned and modified to use an NES-compatible emulator and the relevant validation tools. Because the underlying orchestration logic, session management, and AI conditioning remain unchanged, the cognitive load for developers is significantly reduced. The core challenge shifts from "how do we build a pipeline for a new console?" to "how do we adapt our existing pipeline for a new console?". Following the NES, the same proven pattern can be applied to the PS1, completing the initial three-platform validation. This incremental, iterative approach ensures that each new platform benefits from the lessons learned and the robust architecture refined during the intensive development of the N64 slice.

Synthesis and Actionable Recommendations

This research report has systematically deconstructed the goal of deepening the N64 vertical slice into a fully validated, AI-driven pipeline. The analysis confirms that the most effective path forward is a methodical, phased approach that prioritizes the hardening of the N64-specific workflow before any consideration of cross-platform generalization. The strategy rests on three pillars: the implementation of machine-checkable invariants, the establishment of the `SessionProfile` as the AI's commanding intelligence, and the automation of a complete end-to-end validation pipeline. By treating mathematical formalisms as practical substrates to be validated through code, fuzzing, and CI, the project can ground its ambitious goals in tangible, verifiable reality.

The first pillar, **contract hardening**, involves translating high-level principles of patch safety and hardware budgeting into deterministic Rust validators. The implementation of these checkers for interval containment/disjointness, linear budget inequalities, determinism, and ABI stability provides the necessary guardrails to prevent the AI from making catastrophic errors. These validators are not mere suggestions but hard gates, orchestrated by the `ai_checklist` and enforced by the CI system. Their ability to provide rich, structured diagnostics is what transforms a punitive validation process into a constructive coaching loop for the AI.

The second pillar, **session intelligence**, redefines AI orchestration. By elevating the `SessionProfile` to the status of supreme authority, the system can dynamically control the AI's capabilities based on real-time project health. The AI's available actions are no longer dictated by static text prompts alone but are constrained by the active invariants and CI status encapsulated within the session state. This creates a responsive and adaptive development environment where the AI can autonomously adjust its behavior—from a creative explorer to a focused optimizer—based on empirical evidence from the CI pipeline. The `ci_digest.rs` component is the sensory organ of this system, feeding structured feedback that allows the AI to learn and iterate effectively.

The third pillar, **end-to-end automation**, brings these concepts to life through the `sonia_bridge.rs` and the `.github/workflows/sonia-ai-n64-slice.yml` CI workflow. The bridge acts as the semantic translator, ensuring all AI-generated artifacts conform to the N64 contract before they are ever written to disk. The CI pipeline serves as the automated crucible, executing the entire chain of validation, patching, and testing in a reproducible manner. The successful completion of this cycle, from an AI prompt to a passing build with injected feedback, is the definitive proof-of-concept that validates the entire Sonia model for the N64 platform.

Based on this comprehensive analysis, the following actionable recommendations are proposed:

1. **Prioritize Implementation Order:** Begin by developing the core invariant checkers in `crates/sonia-core/src/invariants/`. Focus on producing rich, structured diagnostic outputs for each. Simultaneously, develop the `sonia_bridge.rs` module to connect the `RomLayout`/`PatchSpec` world to the `ArtifactSpec` world, incorporating the new checkers. Finally, construct the `sonia-ai-n64-slice.yml` CI workflow, starting with stubbed-out tools and progressively replacing them with the real `Starzip` and emulator executables.

2. **Establish the Green Path:** The primary immediate goal is to achieve the first successful execution of the complete end-to-end pipeline. This involves an AI generating a valid `ArtifactSpec`, triggering the CI workflow, and the build passing. This event will serve as the foundational milestone, proving the system works as a cohesive whole.

3. **Implement Rigorous Validation:** Do not rely solely on theoretical correctness. Use a property-based fuzzing framework like LibAFL to generate vast numbers of random inputs for the patch safety and budget checkers, systematically validating

their behavior against edge cases and invalid configurations [41]. Integrate these fuzzing results into the CI pipeline as a mandatory check.

4. **Develop the Session Intelligence Layer:** After the core pipeline is stable, invest effort in enhancing the `SessionProfile` and the `ci_digest` mechanism. Define the specific invariants and execution modes needed for the N64. Implement the logic to translate CI failures into actionable signals within the session state, enabling the AI to adopt context-aware behaviors.

5. **Document Relentlessly:** As each component is built and validated, meticulously document it in `docs/sonia-ai-model.md`. This document should evolve into a comprehensive "flight manual" that captures the contracts, CLI usage, invariant rules, and the complete workflow for the N64 slice. This documentation will be the critical asset for future expansion to the NES and PS1 platforms.

By adhering to this evidence-based, step-by-step plan, the project can systematically mitigate risks, validate its core assumptions through practical implementation, and successfully deepen the N64 vertical slice into a robust, reusable, and AI-orchestrated pipeline, paving the way for a scalable and sustainable platform for AI-driven retro game development.

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